

Unmasking LLM Security: Analyzing Safety Alignment Vulnerabilities via the TwinBreak Methodology (09/06/2025)

[Link](https://arxiv.org/abs/2506.07596): <https://arxiv.org/abs/2506.07596>

Abstract

While Large Language Models (LLMs) have brought transformative benefits to everyday applications, they simultaneously introduce critical security risks. To prevent malicious exploitation, developers rely on safety alignments that automatically reject harmful prompts. However, the emergence of "jailbreak" techniques has proven these defenses are vulnerable. Current methods for studying these bypasses are often bottlenecked by high computational costs, extensive manual effort, or severe degradation of the model's regular utility.

This seminar explores **TwinBreak**, an innovative methodology designed to evaluate and systematically bypass LLM safety alignments. By conceptually treating safety mechanisms as embedded backdoors, TwinBreak identifies and prunes the specific neural parameters responsible for safety blocks. The presentation will detail TwinBreak's novel approach of analyzing intermediate outputs using prompts with high structural and content similarity to precisely isolate safety parameters. Furthermore, the session will introduce the newly developed *TwinPrompt dataset* and review comprehensive experimental findings, demonstrating an 89% to 98% success rate across 16 distinct LLMs with minimal computational overhead. Ultimately, this seminar will provide attendees with a critical understanding of current LLM vulnerabilities and the mechanical nature of AI safety alignment.

LLMs versus the Halting Problem: Characterizing Program Termination Reasoning (26/05/2026)

[Link](https://arxiv.org/abs/2601.18987): <https://arxiv.org/abs/2601.18987>

Abstract

Determining whether a program terminates is a fundamental, undecidable problem in computer science, traditionally addressed by language-specific verification tools that approximate termination but sometimes fail to prove or disprove it. To investigate the extent to which Large

Language Models (LLMs) can reason about this issue, researchers evaluated frontier models on a diverse set of C programs from the SV-COMP 2025 benchmark. The study reveals that models like GPT-5 and Claude Sonnet-4.5, when utilizing test-time scaling, can achieve scores comparable to top-ranked formal verification tools. However, while these models can often correctly infer if a program terminates, their performance degrades as code length increases, and they frequently fail to construct a valid "witness" as a formal proof. This highlights a significant gap between the models' semantic recognition capabilities and their ability to generate symbolic proofs. To better analyze this limitation, the authors introduce a divergence-precondition formulation that characterizes non-termination conditions as logical constraints, aiming to motivate future research into neuro-symbolic verification and broader LLM reasoning on other undecidable problems.

Sci-Reasoning: A Dataset Decoding AI Innovation Patterns (08/01/2026)

[Link](https://arxiv.org/abs/2601.04577) : <https://arxiv.org/abs/2601.04577>

Abstract

The lack of structured data regarding the intellectual processes behind artificial intelligence breakthroughs currently hinders the systematic analysis and development of AI research agents. To solve this, the authors introduce "Sci-Reasoning," the first dataset designed to capture the intellectual synthesis of high-quality AI research by tracing Oral and Spotlight papers from NeurIPS, ICML, and ICLR (2023–2025) to their key predecessors using an LLM-accelerated pipeline. Their analysis of this structured data uncovered 15 distinct thinking patterns, with three dominant strategies accounting for 52.7% of the approaches: Gap-Driven Reframing (24.2%), Cross-Domain Synthesis (18.0%), and Representation Shift (10.5%). The study notes that the most powerful innovations often combine these patterns, providing actionable "innovation recipes" that enable quantitative studies of scientific progress and offer structured reasoning trajectories to train the next generation of AI research agents.